MILITARY SPECIFICATION

CONTAINER, ENVIRONMENTAL PROTECTION MEDICAL SUPPLIES

This specification is approved for use by Defense Personnel Support Center, Department of Defense, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Container, Environmental Protection, Medical Supplies designed to prevent medical supplies from freezing in Arctic type environments.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents where listed in the contract form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-527	- Enamel, Alkyd, Lusterless.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-I-24172	- Insulation, Plastic, Cellular Polyurethane, Rigid, Preformed and Foam-In-Place.
MIL-F-45216	- Foam-In-Place Packaging, Procedures for.

STANDARDS

Beneficial comment (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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MIL-STD-129 - Marking for Shipment.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-171 - Finishing Metal and Wood Surfaces.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of
MIL-STD-810 - Environmental Test Methods.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

DRAWING NO. 25509 - Container, Environmental Protection, Medical Supplies.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein.

Technical Manual of the United States Army Medical Bioengineering Research and Development Laboratory.

(Application for copies should be addressed to Commander United States Army Medical Bioengineering Research and Development Laboratory, Fort Detrick, Frederick, MD 21701-5010.)

3. REQUIREMENTS

3.1 First article When specified in the procurement document, the contractor shall furnish preproduction sample unit(s) for Article inspection and approval (see 4.5, 6.2 and 6.4).

3.2 Materials.

3.2.1 Aluminum and aluminum alloys

3.2.1.1 Aluminum extrusions, 6063 alloy. The extrusions (bar, rod, shapes, tube, and wire used in the construction) shall be made out of 6063 aluminum alloy.

3.2.1.2 Aluminum plate and sheet, 5052 alloy. The aluminum plate and sheet (used in the construction) shall be made out of 5052 aluminum alloy.

3.2.1.3 Aluminum castings, 356 alloy. Aluminum alloy castings (used in the construction) shall be 356 aluminum alloy.

3.2.2 Corrosion resisting steel (CRES), stainless steel. The material shall be properly passivated.

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3.2.2.1 Corrosion resisting steel bars, wire, shapes, and forgings. Corrosion resisting steel bars, wire, shapes, and forgings shall be used in the construction of the items.

3.2.2.2 Corrosion resisting steel, plates, sheets, and strip. Corrosion resisting steel, plates, sheets, and strip shall be used in the construction of the items.

3.2.3 Hardware and fasteners. All hardware and fasteners used in the construction of the items shall be corrosion resisting material or have a corrosion resistant finish.

3.2.3.1 Machine screws, corrosion resisting steel. All machine screws used in the construction of the items shall be corrosion resistant steel.

3.2.3.2 Self-tapping screws, corrosion resisting steel. All self-tapping screws used in the construction of the items shall be corrosion resisting steel screws and all screws must be AISI 300 series stainless steel.

3.2.3.3 Rivets, aluminum, solid and blind, pull-stem. All rivets used in the construction of the items shall be aluminum.

3.2.4 Insulation, polyurethane foam, rigid. The rigid polyurethane foam insulation used in the construction of the items shall meet the requirements (material, color, odor, uniformity, property values, cure temperature, component ratio, and workmanship) of MIL-I-24172 for type II-foamed-in-place, class 1-1.7 to 2.5 pounds per cubic foot (density). Fire retardancy is not required. See 3.6.2 (of this specification) for the components required to produce the foam.

3.2.5 Rubber.

3.2.5.1 Rubber, handle grip. The rubber handle grip (used on the lifting handle) shall be extruded tubing.

3.2.5.2 Rubber, silicone, lid seal gasket. The lid seal gasket shall be fabricated from silicone rubber.

3.2.6 Plastic.

3.2.6.1 Plastic, thermosetting, phenolic. Phenolic resin shall be used to mold the control box and lid used for the temperature control and alarm control and shall be impact resistant phenolic resin.

3.2.6.2 Plastic, glass reinforced. Polyester glass-mat sheet laminates shall be used in the construction.

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3.3 Design and construction. The environmental container shall be insulated, electrically heated, and temperature controlled and shall be designed to protect medical supplies from freezing during transport and storage in an Arctic environment when fabricated and constructed according to Drawing No 25509 in section 2 of this specification.

3.3.1 Mechanical design and construction. The environmental container shall be a rigid foam insulated box, top loading with aluminum interior and exterior shells. A recessed handle with a two-hand grip bar for mittened hands shall be provided on each of the four sides to allow personnel to lift and move the environmental container. Skids shall be provided to protect the bottom of the exterior shell and to facilitate forklift handling of the environmental container. Latches shall be provided to insure compression of the lid seal gasket. A double seal gasket shall be recessed into the lid of the chest. This will prevent damage to the gasket and minimize the chance of trapping moisture and freezing the seal to the sealing surface. The fiberglass plastic strips (used under the seal and as the sealing surface) shall act as thermal breaks in the heat transfer path between the inner shell and outer shell. A hasp and padlock shall be used to secure the contents. The lid and skids shall have male and female dimples to act as keys for stacking of one environmental container on another.

3.3.2 Electrical/electronic design and construction. The environmental container shall employ an electrical system to protect medical supplies subject to damage by freezing in an Arctic environment. This system shall include power cables, resistance heaters and temperature control, alarm control, and remote visual/audible ralarm.

3.3.2.1 Electrical power sources. The electronic system of the environmental container shall maintain and control the storage temperature and meet the performance requirements of this specification when powered by forward combat area power sources producing 28 volt direct current (DC) or 120 volt single phase alternating current (AC) at 50 Hertz, 60 Hertz, or 400 Hertz frequency. These power sources shall include engine driven electrical generator sets, military vehicle 28 volt DC electrical systems, and military aircraft 28 volt DC or 120 volt AC electrical systems.

3.3.2.1.1 Electrical power, engine driven generator sets. The electrical system shall function and perform properly when engine driver generator sets are used as sources of electrical power.

3.3.2.1.2 Electrical power, military vehicle, 28 volt DC. The electrical system shall function and perform properly when military vehicle 28 volt DC electrical systems are used as sources of electrical power.

3.3.2.1.3 Electrical power, military aircraft. The electrical system shall function and perform properly when military aircraft 28 volt DC, 120 volt single phase AC, 60 Hertz and 400 Hertz electrical systems, are used as sources of electrical power.

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3.3.2.2 Power cables. Shall provide power cable and power supply adapter cable (pig tails) assemblies per environmental container. The cable and wire shall be light duty, low temperature, flexible cable, and waterproof cable (including connections and adapters). The power cable and power supply adapter cables shall be identified by banding using plastic band, heat shrinkable tubing, or aluminum band Each wire on the power supply adapter cable shall be banded with and identified for proper connections to a power supply such as +, -, and/or ground. The 120 volt AC adapter cable shall be able to be converted to commercial use by connecting it to a a three-prong power plug. The cable assemblies shall be interchangeable and meet the dimensions, tolerances, and requirements of Drawing No. 25509.

3.3.2.3 Temperature control construction. The temperature control shall sense and control the material storage temperature through the use of a snap action, fixed set point, precision thermostat that turns four resistance type strip heaters on and off as required to keep the freezable portion of the medical supplies stored in the chest from freezing. The electrical circuit shall automatically connect the heaters in series for 120 volt AC operation or in parallel for 28 volt DC operation. The strip heaters, temperature control, and thermostat shall be easily replaceable. The heaters shall be designed to be interchangeable between environmental containers.

3.3.2.4 Temperature control construction. The temperature control circuit shall be contained on a printed circuit board (PCB) housed within an aluminum electronics box on the inside wall of the environmental container. The temperature control circuit shall be mounted in the upper part of the electronics box and shall be ?r?rremovable (remove two screws and an electrical edge-connector) for servicing or rreplacing. Failed control circuit boards shall be replaced (rather than repaired) under field conditions. Wiring harness (embedded in the insulation between the container's inner and outer wall) shall be used to connect strip heaters to the control circuit board. The heaters shall be housed behind metal strips (on the inside wall of the environmental container) and shall be replaceable (remove terminal board connection). Both thermostats (that govern heater operation) shall be mounted to the environmental container inner wall (below the electronics box). Both thermostats shall be connected to embedded wiring running to the terminals of the edge connector for the control circuit board. Fuses (for the electrical circuits) shall be mounted in the battery fuse box (above the electronics box). There shall be one fuse for the 28 volt DC circuit and the 120 volt AC circuit and a spare fuse for each circuit mounted in an adjacent fuse clip.

3.3.2.5 Temperature controls, electrical circuit design. See equipment description and data (section II), technical principles of operation (section III), and schematic diagram in the Technical Manual of the United States Army Medical Bioengineering Research and Development Laboratory specification.

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3.3.2.6 Alarm control design. The chest alarm shall sense the internal temperature of the environmental container and shall activate an audiovisual alarm when the temperature falls below 35°F (1.7°C). The audiovisual alarm shall be housed in a separate small container that plugs into the chest and shall be collocated with the chest or positioned 20 feet away. The alarm box shall plug into either of two standard 1/4-inch telephone jacks located in the rear handle recess of the environmental container. The second jack (slave) shall allow the interconnection of the alarm control circuits from several chests so that one alarm shall monitor all the environmental containers.

3.3.2.7 Alarm control construction. The alarm control circuit shall be contained on the lower control circuit board inside the main electronics box.

The temperature sensing shall be performed by a thermistor (mounted directly on the circuit board) that triggers the external alarm through an operational amplifier. The sensing/alarm circuit shall be powered from the same 120 volt AC/28 volt DC source that powers the heater circuits. In the event of power failure or long-term interruption, there are two 9-volt nickel cadmium (NiCAD) rechargeable batteries (located on the inside cover of the fuse box) that shall take over as a power source for the alarm circuit. The batteries shall be automatically charged from the alarm circuit during normal operation of the environmental container from an external power source.

3.3.2.8 Remote alarm construction. The remote alarm shall be a lightweight [illegible] box with a red light and alarm. When the alarm is activated, an electronic beep shall sound and the red light shall flash. The remote alarm box shall include 20 feet of electrical cable which shall be stored inside the box (with the remote alarm box) when not in use. The remote alarm box, cable, and plug shall function interchangeably with all the environmental containers.

3.3.2.9 Alarm control electronic circuit design. See equipment description and data (section II), technical principles of operation (section III), and schematic diagram in the Technical Manual of the United States Army Medical Bioengineering Research and Development Laboratory.

3.3.2.10 Thermostats. There shall be two thermostats (hermetic) within the environmental container, one shall be for high temperature control and the other for low temperature control.

3.3.2.11 Thermometer. The thermometer (located in the front handle recess of the environmental container) shall be a dial thermometer and shall measure the internal temperature with a range of -40° to 120°F.

3.3.2.12 Heaters. There shall be four strip heaters per environmental container. One strip heater shall be mounted per inside wall.

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3.3.2.13 Spacer strips. Spacer strips shall be located inside the environmental container along the walls and base. The spacer strips shall be insulating strips used to minimize contact of the stored supplies with the environmental container walls.

3.4 Dimensions. The dimensions and tolerances of the items included in this specification shall conform to the drawing listed.

3.5 Function and performance.

3.5.1 Operational environment. The environmental container shall operate satisfactorily in Arctic environments and shall meet the requirements of MIL-STD-810 for low temperature (Basic Cold (C1)), Cold (C2), and Severe Cold (C3)) method 502.2, temperature shock method 503.2, salt fog (corrosion resistance) method 509.2, vibration (basic transportation, large assembly transport, loose cargo transport, helicopter, and ground mobile) method 514.3, and icing/freezing rain method 521.0.

3.5.2 Function. The lid, lid brace, latches, and hasp shall allow opening, closing, sealing, and locking the lid with no binding of the moving parts. The stacking dimples shall key two or more environmental containers when stacked one on top of the other. The handles shall extend for use and retract when released without binding. The alarm box shall plug into and be unplugged from its connector without binding or the use of more than 5 pounds of force. The environmental container shall hold 125 pounds of medical items. The lid latches shall pull the lid down to the box sufficiently to assure contact of one lid seal with the box on the total length of that seal and at least 80 percent of the second seal.

3.5.3 Performance.

3.5.3.1 Performance with power. The environmental container shall maintain the freezable portion of a standard 125-pound test load between 35°F (1.7°C) and 45°F (7.2°C) with an external environment corresponding to the relative humidity for Climatic Design types: Basic Cold (C1), Cold (C2), and Severe Cold (C3) as described in MIL-STD-810, which includes temperatures from -5°F (-20.6°C) down to -60°F (-51°C), relative humidities at saturation, and wind speeds of 16 feet per second. Refer to Quality Assurance Provisions (see 4. through 4.8.3.1) for test method.

3.5.3.2 Performance without power. The environmental container shall maintain the freezable portion of a 125-pound load unfrozen at or above 32°F (0°C) for 8 hours after the power has been disconnected. The load shall have been between 40°F (4.4°C) and 45°F (7.2°C) before being disconnected using the environmental container electrical heating and temperature control system. The ambient environment for this requirement shall include temperature at -50°F (-45.6°C), relative humidity at saturation, and wind speed of 16 feet per second as in the Severe Cold (C3) Climatic Design Type in MIL-STD-810. Refer to Quality Assurance Provisions (see 4. through 4.8.3.1) for test method.

3.6 Fabrication.

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3.6.1 Welding Welding shall be durable and sound in every detail.

3.6.1.1 Cleaning aluminum alloy. All aluminum alloy shall be cleaned not more than 48 hours before welding, spot, and arc.

3.6.1.2 Fastening, alternate processes. The spot weld and rivet assemblies specified on the drawings may be replaced by alternate methods of construction, provided the contractor furnishes proof of equivalency to the contracting officer prior to use of the alternate method.

3.6.2 Foam-in-place process The foam-in-place process shall be either aerosol or machine dispensing for type V-special designs, Class 1-in-place, style 1-complete fill as described in MIL-F-45216. Foam-in-place process shall be as described in MIL-F-45216. If there is any conflict between MIL-I-24172 and MIL-F-45216, MIL-I-24172 shall govern except for thickness and thickness tolerance, float, bulging, buckling, and distortion of the inside and outside surfaces of the container shall be governed by the drawing of the foamed assembly. Fire retardancy shall not be required.

3.6.2.1 Pretreatment of aluminum. The aluminum sheet shall be cleaned and surface treated. After the aluminum has been cleaned and treated, the inner surfaces of the environmental container (in contact with the polyurethane foam insulation) shall be coated within 4 hours before foaming with 3M panel adhesive No. 2318. Any substitution for No. 2318 panel adhesive must be approved by the contracting officer.

3.6.2.2 Positioning blocks Blocks used to position the inner shell in the outer shell (during foam-in-place operations) shall be polyurethane, rigid, type I, class I, and shall meet the requirements of MIL-I-24172.

3.6.2.3 Approval of foam materials, process, and equipment. Contracting officer must approve foam materials, process, and equipment prior to beginning foam-in-place production operations. The approval shall be based on the results of actual tests in the contractor's facility. The materials, equipment, and process shall be the same as or equal to what will be used in production. The tests shall conform to test requirements in the material specification for MIL-I-24172.

3.7 Weight and volume The environmental container assembly (without power cable and power supply adapter cables) shall not weigh more than 100 pounds. The net internal volume of the environmental container shall be 6.0 cubic feet.

3.7.1 Weight and volume, foam. The insulated volumes of the lid and the container shall be 1.6 cubic feet and 6.4 cubic feet, respectively. The weight of polyurethane foam insulation shall be 3.1 pounds for the lid and 12.8 pounds for the container.

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3.8 Finish and color. The environmental container shall be finished (inside/outside) for type I (exposed) and shall include such climatic elements as temperature extremes, humidity extremes, rain, hail, snow, sleet, and salt laden air and shall meet the requirements of MIL-STD-171, appendix A. The interior and exterior surfaces shall be finished by cleaning, wash priming, and epoxy priming according to MIL-STD-171.

3.8.1 Color. The interior and exterior surface shall be painted forest green and shall meet the requirements of MIL-STD-171.

3.8.2 Film thickness. The pretreatment and primers shall have the proper film thickness after the paint has dried. Refer to 5.2.3 of MIL-STD-171 for the proper film thickness. Film thickness for camouflage shall be a minimum of 1.6 mils as required in MIL-STD-171.

3.8.3 Application of camouflage paint. The application of camouflage paint shall meet requirements of MIL-STD-171 for paint thickness, dust coat, and dry time before applying the second coat. Camouflage pattern painting when specified in the contract shall be in addition to the requirements of this specification and shall conform to the pattern stipulated.

3.8.4 Impregnation of castings. After cleaning and before surface treatment with a wash primer, the castings shall be properly impregnated.

3.8.5 Cleaning and preparation of plastics for painting. The molded phenolic control boxes and fiberglass plastic strips used to connect the inner and outer aluminum shells shall be solvent wiped to remove any waxes, mold releases, oils, or soluble foreign substances and shall be scuff sanded with 180 grit sandpaper. The sanding should be solvent wiped a second time to remove sanding dust and any waxes, mold releases, oils, or foreign substances on the surfaces to be painted. Do not sand blast.

3.8.6 Surfaces not to be painted. The silicone rubber seals, power cables, and power supply adapter cables shall not be painted.

3.8.7 Blasting. Blasting shall not be used on surfaces that could be damaged by blasting, such as machined parts and sheet metal thickness less than 0.100 inch.

3.8.8 Stenciling and marking. The stenciling and markings shall be made with a lusterless enamel and shall be applied in lusterless white No. 37875 (see table I of TT-E-527), on forest green painted surfaces.

3.8.9 Painting after foaming. For all painting procedures after foaming of the lid and the environmental container, the maximum temperature of the aluminum inner and outer shells (the paint) shall be limited to 125°F (51.7°C).

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3.9 Identification and marking. Each environmental container shall have a permanently fastened identification plate. The information on the plate shall include the following contract number, name or registered trademark of manufacturer, manufacturer's code number, design activity code number, US (Government ownership designation), item name, National Stock Number (NSN), and serial number. All lettering on the identification plate is to be at least 3/16 inch (.187 inch) high. The plate shall be fastened to the container portion of the chest assembly.

3.9.1 Cautions and instruction. All cautions and instruction shall be stenciled or screened on the environmental container in white lusterless enamel as required in TT-E-527. See drawing No. 25509 for the legend and location. All letters shall be at least 3/16 (.187) inch high.

3.10.1 Workmanship, foam-in-place insulation. The foam-in-place shall meet the requirements of MIL-I-24172 for uniformity in quality and serviceability, free from defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor shall be responsible for the performance of all inspection requirements according to this specification. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements according to this specification unless it is not approved by the Government. The Government shall reserve the right to perform any of the inspections set forth in the specification where such inspections shall be deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records shall allow identification and traceability of the lot number of raw materials used and of manufacturing and assembly processes, including final packaging. Records of inspections, examinations, and tests performed by or for the contractor specified in this section shall be maintained by the contractor and made available to the Government, upon the Government's request, during the performance of the contract and for a period of 3 years after delivery of the products.

4.1.2 Inspection. Inspection shall be defined as both examination (such as visual, dimensional, and functional investigation without the use of special laboratory appliances or procedures) and testing (determination by laboratory and technical investigation, destructive and non-destructive, the physical, electrical and chemical properties) of materials, components, product, and packaging to determine compliance with this specification.

4.1.3 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The contractor shall establish and maintain a calibration system to control the accuracy of the measuring and test equipment, and the contractor's calibration system shall be acceptable to the Government.

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4.2 Classification of inspections. The inspections specified shall be classified as material inspection, first article inspection, or quality conformance inspection.

4.3 Materials inspection. Material inspection shall consist of certification supported by verifying data that the materials used in fabricating the environmental containers are in accordance with applicable referenced specifications or requirements prior to fabrication.

4.4 Inspection conditions. All inspections shall be performed in accordance with test conditions.

4.5 First article inspection. First article inspection shall be performed by the contractor, after the award of the contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on preproduction sample units that have been produced with equipment and procedures normally used in production. First article approval shall be valid only on the contract under which it is granted unless extended by the Government to other contracts.

4.5.1 Sample size. At least two environmental containers shall be subjected to first article inspection.

4.5.2 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall be before subassembly, after subassembly (before foaming), after subassembly of shells and lids (before foaming), after foaming (before final assembly), after final assembly and painting, after final assembly (performance with power), after final assembly (performance without power), and after final assembly (weight assembled and painted).

4.6.1.1 Inspection lot. An inspection lot shall consist of all environmental containers covered by this specification produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and may resubmit for reinspection. Resubmitted lots shall be reinspected. The level and type of inspection are to be determined by responsible authority. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.7 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the container specification and the marking requirements of MIL-STD-129.

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4.8 Methods of inspection.

4.8.1 Examination. Examine representative samples of materials, components, and product to determine compliance with the requirements, specifications, and standards specified.

4.8.2 Testing. Test representative samples of materials, components, and product to determine compliance with requirements, specifications, and standards specified.

4.8.2.1 Low temperature performance test with power. The environmental container shall be tested at the low temperature extreme of -60°F (-51°C), relative humidity at saturation, and with air speeds of 16 feet per second as required in MIL-STD-810, method 502.2. See paragraph 3.5.3.1 of this specification for requirements. See appendix A for a list of items required to make up the 125-pound test load. The load temperatures shall be monitored and recorded at locations listed in appendix B at least every 12 minutes until the temperature of the load stabilizes for at least 2 hours. All items in the load shall be between 35°F (1.7°C) and 45°F (7.2°C) when placed in the environmental container or when the test is started. Commercial power (120 volts, single phase, 60 Hertz, AC) shall be used to operate the electrical system. At least one environmental container shall be tested using 120 volts, single phase, 400 Hertz, AC power. At least one environmental container shall be tested using 28 volts DC power. At least three environmental containers shall be tested. The environmental container shall fail if the temperature of any item in the chest falls below 35°F (1.7°C).

4.8.2.2 Low temperature performance test, without power. The environmental container shall be tested at the low temperature extreme of -50°F (-45.6°C), relative humidity at saturation, and air speeds of 16 feet per second. The 125-pound test load shall have been stabilized between 40°F (4.4°C) and 45°F (7.2°C) in the environmental container electrical heating and temperature control system. See appendix A of this specification for a list of items required for the 125-pound test load. The load temperatures shall be monitored and recorded at locations listed in appendix B while the load is stabilizing and during the test with power. See paragraph 3.5.3.2 on performance without power in this specification. After the test load has been stabilized for at least 2 hours, the power shall be disconnected and the temperature of the load shall be monitored and recorded every 6 minutes for at least 8 hours. The operation of the alarm, chest mounted and remote, shall be recorded. The test shall follow the applicable procedures in MIL-STD-810. If an item in the load freezes, develops ice crystals within the item, or is irreversibly damaged by temperatures of 32°F (0°C) or below, the environmental container shall fail the test without power. At least three environmental containers shall be tested without power.

4.8.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual testing by the contractor, provided lot identity has been maintained and shall be demonstrated to the Government and provided also that the test methods and results obtained shall comply with those described in this specification. The certificate

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shall include the name of the contractor, contract number, name of the manufacturer or supplier, manufacturer's identification of the component or the material, name of the component or material, lot number, lot size, sample size, date of testing, test method, and individual test results.

4.8.3.1 Polyurethane certificates of quality. Certificates of quality for polyurethane materials are not acceptable in lieu of specific testing required

5 PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. Each environmental container shall be clean dry and have the lid closed but not latched to minimize compression of the seal. The chest shall be further packaged in accordance with MIL-P-116, method III.

5.1.2 Level C. The level C preservation for chest shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing.

5.2.1 Level A. Each environmental container shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. The environmental container shall be covered on the top, ends, and sides with a 3 mil plastic shroud. The front end of the environmental container containing the latches shall be cushioned with fiberboard pads or other suitable cushioning material to prevent rupture of the exterior container. A 3/4 inch piece of plywood shall be placed on the interior bottom of the container in order that the weight of the environmental container shall be evenly distributed over the entire bottom of the fiberboard box. Closure, banding, and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Each environmental container shall be packed as specified (see 5.2.1), except that the exterior container shall conform to PPP-B-636, class domestic and waterproofing shall be required.

5.2.3 Level C. The level C packing for environmental containers shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

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5.2.4.1 Levels A and B. Environmental containers packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Environmental containers packed as specified in 5.2.2 shall be unitized as specified in MIL-STD-794.

5.3 Marking

5.3.1 Levels A, B, and C. Each exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.

6 NOTES

6.1 Intended use. The environmental container will be used by front-line medical units in ar Arctic environment equivalent to Climatic Design Types Basic Cold (C1), Cold (C2), and Severe Cold (C3), including temperatures between -5°F (-21°C) to -60°F (-52°C). The environmental container will be used to support daily operations, tactical moves, depot-to-unit shipments, and for warehouse and outside storage of medical supplies subject to damage by freezing.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following.

a. Title, number, and date of this specification.

b. National Stock Number

c. Selection of packaging.

d. When a first article is required

e. Color of finish and camouflage pattern, if required.

6.3 This specification covers the following item listed in the Federal Supply Catalog.

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NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6530-01-192-9459	Container, Environmental Protection, Medical Supplies, 29.25" x 29.25" x 30.84"

6.4 First article. When a first article is required, it will be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article will be preproduction samples. The first article will consist of two units. The contracting officer will include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

6.5 Availability of color chips. For camouflage paints, color chips can be obtained from Organic Chemicals and Coating Research Group, Materials Technology Lab, DRDME-VO, USAMERADCOM, Fort Belvoir, VA 22060. It should be noted that acceptance of an end item will not be based specifically on color when camouflage paints are used but rather on whether the paint was approved by the US Army Mobility Equipment Command, Attn DRDME-VO, Fort Belvoir, VA 22060. Where color is specified, color chips are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.6 Operator and organizational maintenance manual. The contractor will furnish two copies of the Operator and Organizational Maintenance Manual describing complete operation and service/maintenance instructions for each environmental container (see Technical Manual of the United States Army Medical Bioengineering Research and Development Laboratory). Manuals will be furnished in accordance with Data Item Descriptions 0005 and 0006 of DPSC-DEPMEDS-AT(DM). The contractor will agree to a limited copyright release to the military services to reproduce all material furnished under this contract for use in military services training program. The manual will include a typed sheet listing all the parts and equipment and the location of such parts and equipment. The manual will include repair parts list, if any, which the Government elected to purchase. The contractor will submit two initial copies to the Contracting Officer for comment/approval prior to final printing and issuance

Preparing Activity
DLA-DM

Project No. 6530-1718

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APPENDIX A

125-POUND LOAD FOR LOW TEMPERATURE TEST

Number of items	Weight (pounds)	Description
16	56	100 milliliters (ml) of 5 percent saline solution in glass bottle
5	46	1 gallon of water in plastic jug
20	20 5	500 ml of water and citrate solution in plastic blood bag
1	2.5	1 quart of water in plastic bottle
	125	pounds of total load

Stacking Method Bottles will be protected on all sides, top, and bottom with corrugated cardboard to prevent breakage. Bottles will be stacked on corrugated cardboard in bottom of environmental container, gallon jugs placed on top of bottles, a square of corrugated cardboard placed on top of gallon jugs, and blood bags laid on top of corrugated cardboard square.

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APPENDIX B

THERMOCOUPLE LOCATIONS

Number	Location
1	Side of 1000-milliliter (ml) saline bottle at bottom corner of chest
2	Inside gallon plastic container at geometric center of load
3	On the bottom surface of 500-ml blood bag on top of load
4	On heater cover halfway between top and bottom of chest
5	Center of lid, interior surface
6	Center of bottom interior surface of chest
7	Bottom of 1000-ml saline bottle
8	Airspace between plastic containers near the center of the load
9	Corner of interior surface halfway between top and bottom, not at corner next to controls
10	Center of airspace above load

Location H1-4-25

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-37964	Container, Environmental Protection Medical Supplies

3a NAME OF SUBMITTING ORGANIZATION

b ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional

b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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